Example 10.5

Solve the following LPP

Maximize $Z = 2x_1 + 5x_2$ subject to the conditions $x_1 + 4x_2 \leq 24$ $3x_1 + x_2 \leq 21$ $x_1 + x_2 \leq 9$ and $x_1, x_2 \geq 0$

Solution:

First we have to find the feasible region using the given conditions.

Since both the decision variables x_1 and x_2 are non-negative, the solution lies in the first quadrant.

Write all the inequalities of the constraints in the form of equations.

Therefore we have the lines $x_1 + 4x_2 = 24$; $3x_1 + x_2 = 21$; $x_1 + x_2 = 9$

$x_1 + 4x_2 = 24$ is a line passing through the points (0 , 6) and (24 , 0).

[(0,6) is obtained by taking $x_1 = 0$ in $x_1 + 4x_2 = 24$, (24 , 0) is obtained by taking $x_2 = 0$ in $x_1 + 4x_2 = 24$].

Any point lying on or below the line $x_1 + 4x_2 = 24$ satisfies the constraint $x_1 + 4x_2 \leq 24$.

$3x_1 + x_2 = 21$ is a line passing through the points (0, 21) and (7, 0). Any point lying on or below the line $3x_1 + x_2 = 21$ satisfies the constraint $3x_1 + x_2 \leq 21$.

$x_1 + x_2 = 9$ is a line passing through the points (0 , 9) and (9 , 0) .Any point lying on or below the line $x_1 + x_2 = 9$ satisfies the constraint $x_1 + x_2 \leq 9$.

Now we draw the graph.

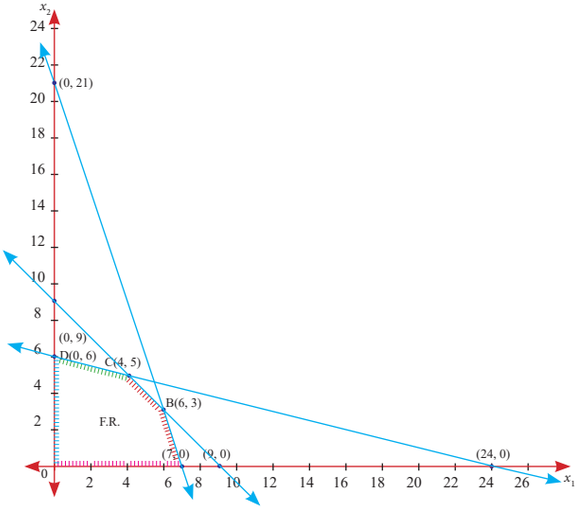


Fig 10.1

The feasible region satisfying all the conditions is OABCD. The co-ordinates of the points are O(0,0) A(7,0);B(6,3) [the point B is the intersection of two lines $x_1 + x_2 = 9$ and $3x_1 + x_2 = 21$];C(4,5) [the point C is the intersection of two lines $x_1 + x_2 = 9$ and $x_1 + 4x_2 = 24$] and D(0,6).

Corner points	$Z = 2x_1 + 5x_2$
O(0,0)	0
A(7,0)	14
B(6,3)	27
C(4,5)	33
D(0,6)	30

Table 10.2

Maximum value of Z occurs at C. Therefore the solution is $x_1 = 4$, $x_2 = 5$, $Z_{max} = 33$.

Example 10.6

Solve the following LPP by graphical method Minimize $z = 5x_1 + 4x_2$ Subject to constraints $4x_1 + x_2 \geq 40$; $2x_1 + 3x_2 \geq 90$ and $x_1, x_2 \geq 0$

Solution:

Since both the decision variables x_1 and x_2 are non-negative, the solution lies in the first quadrant of the plane.

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